

Smart Grid and Demand Response Technologies

1. Recommendations and Conclusion

The global electric grid is undergoing its most profound transformation since its inception in the late 19th century. The integration of distributed energy resources (DERs), proliferation of smart metering infrastructure (AMI), rapid growth of intermittent renewable generation, and emergence of transactive energy markets are collectively redefining how electricity is generated, transmitted, distributed, and consumed. Smart grid technologies and demand response programs sit at the nexus of this transformation, enabling bidirectional communication, dynamic pricing, and flexible load management at unprecedented scale.

The U.S. smart grid investment has exceeded \$70 billion since 2009, with the DOE estimating that full smart grid deployment could reduce electricity consumption by 12% and peak demand by up to 20%, avoiding \$170 billion in generation, transmission, and distribution infrastructure costs by 2030. Demand response (DR) programs currently enrolled approximately 38 GW of dispatchable demand-side resources in North America as of 2024, a figure expected to grow to 80 GW by 2030.

This whitepaper examines smart grid architectures, communication protocols, demand response program structures, grid-interactive building technologies, virtual power plant (VPP) platforms, and the emerging transactive energy paradigm. It provides performance benchmarks, implementation guidance, case studies, and a forward-looking assessment of AI-driven grid optimization and the role of distributed flexibility in supporting 80% renewable electricity targets.

The traditional power grid was designed around the physics of centralized, dispatchable generation flowing in one direction to passive consumers. This paradigm is increasingly incompatible with a power system characterized by: (1) 15% of U.S. electricity from variable wind and solar (on track for 30% by 2030); (2) 3.5 million distributed solar installations injecting power back into distribution circuits; (3) 6 million electric vehicles creating new, large, and unpredictable load centers; and (4) consumer expectations for real-time pricing information and automated energy management.

Grid operators face mounting challenges managing system stability with increasing shares of non-dispatchable generation. Frequency regulation — maintaining grid frequency within 59.95–60.05 Hz — requires continuous balancing of generation and load. Historically provided by spinning reserve from thermal generators, frequency response must increasingly come from demand-side flexibility, battery storage, and demand response as thermal capacity retires. NERC's 2024 Long-Term Reliability Assessment identifies demand response and storage as critical

reliability resources for 12 of 24 regional assessments.

A smart grid architecture spans the full electricity value chain from generation to end-use, integrating three fundamental layers: (1) the physical energy infrastructure (generation assets, transmission lines, distribution feeders, substations, and customer premises equipment); (2) the communication and sensing layer (Advanced Metering Infrastructure, smart sensors, SCADA systems, and IoT devices); and (3) the data analytics and control layer (energy management systems, demand response management systems, distributed energy resource management systems, and market platforms).

Advanced Metering Infrastructure encompasses the complete system of smart meters, communication networks, and data management systems that enable two-way data exchange between utilities and customers. As of 2024, 115 million smart meters are installed in U.S. homes and businesses, representing 75% of total electricity accounts. AMI enables hourly or sub-hourly consumption data (most utilities: 15-minute intervals), remote connect/disconnect, outage detection, power quality monitoring (voltage, current, power factor), and time-of-use pricing signals.

Smart meter communication architectures use multiple wireless technologies: RF mesh networks (Itron OpenWay, Landis+Gyr Gridstream) operating in 902–928 MHz ISM band with typical range of 300–500 meters between meters; cellular (LTE-M, NB-IoT) for low-latency applications; and PLC (Power Line Communication) in densely populated areas. AMI data generates 1–2 GB per day per 100,000 meters, requiring robust head-end systems and data analytics platforms capable of processing terabytes of consumption data for billing, load forecasting, and demand response coordination.

Demand response programs are classified by their dispatch mechanism, customer incentive structure, and operational timeline. The Federal Energy Regulatory Commission (FERC) Order 2222 and Order 745 define the regulatory framework for DR participation in wholesale markets, establishing compensation equivalency between demand reduction and supply-side resources.

OpenADR (Open Automated Demand Response) is an open-source communications data model and protocol developed by the Lawrence Berkeley National Laboratory and standardized as ASHRAE/IEC 62746-10. OpenADR 2.0b enables fully automated DR execution without human intervention, communicating price signals (Real-Time Pricing events), reliability events (DR events), and grid condition signals from a Virtual Top Node (VTN, operated by utility/aggregator) to Virtual End Nodes (VEN, in customer facilities).

OpenADR event signals include: simpleLevel (0–3 indicating normal/moderate/high/special pricing); price (actual \$/kWh or \$/kW signal); load control (explicit kW reduction target); and eMIX pricing curves. Facility BEMS or EMS platforms receiving OpenADR signals automatically execute pre-programmed response strategies: raising thermostat setpoints by 2–4°F, shifting EV charging to off-peak, cycling non-critical loads, and dispatching on-site generation or storage. Response automation eliminates the 15–40% non-response rate observed in manually-dispatched programs.

A Distributed Energy Resource Management System (DERMS) is the operational platform used by utilities and grid operators to monitor, forecast, and dispatch aggregated DER fleets at distribution system level. Unlike traditional SCADA which monitors bulk transmission, DERMS provides visibility and control of solar PV, battery storage, EV chargers, demand response, and flexible loads connected to distribution feeders. Leading DERMS platforms (Smarter Grid Solutions ANM, GE Grid Solutions ADMS, Oracle Utilities Network Management) process real-time data from tens of thousands of DER units simultaneously.

DERMS functional requirements include: (1) real-time DER monitoring (< 5 second latency); (2) hosting capacity analysis for DER interconnection approval; (3) volt/VAR optimization for voltage management with high solar penetration; (4) constraint management to prevent feeder overloads; and (5) market interface for DER participation in wholesale energy and ancillary service markets. FERC Order 2222 (effective July 2021) mandates that RTOs/ISOs allow DER aggregations to participate in all capacity, energy, and ancillary service markets.

Grid-Interactive Efficient Buildings (GEB) are defined by the DOE as buildings that use smart technologies and on-site DERs to optimize energy use for both building users and the grid simultaneously. The GEB framework encompasses four grid services: (1) Efficiency — continuous reduction of energy consumption; (2) Load Shed — reducing peak demand on utility request; (3) Load Shift — moving flexible loads to favorable grid periods; and (4) Modulation — providing rapid, automated grid services (frequency regulation, operating reserves).

GEB value stacking — simultaneously capturing energy, demand charge, DR incentive, and grid service revenue streams — can deliver total annual value of \$0.15–\$0.55/sq ft in commercial buildings. A 500,000 sq ft office building participating in all four GEB service categories could generate \$75,000–\$275,000 in annual combined savings and revenue, representing a 40–80% improvement over single-program participation. Building thermal mass, battery storage, flexible HVAC, and EV charging are the primary GEB assets.

Commercial and industrial customers participating in wholesale electricity markets follow a structured workflow aligned with market clearing timelines. In PJM (the largest RTC in North America, serving 65 million customers), the day-ahead market closes at 12:00 PM for next-day scheduling. Demand response aggregators submit demand resource bids specifying available MW reduction, minimum notification time, and minimum event duration by 12:00 PM. Cleared DR resources receive binding dispatch instructions for the following day.

Real-time markets (5-minute clearing intervals in most RTOs) provide additional participation opportunities for facilities with automated response capability. Load aggregators with DERMS platforms can provide 5-minute automated adjustments to meet real-time dispatch instructions, capturing real-time LMP prices that can reach \$500–\$2,000/MWh during scarcity events. In the 2024 polar vortex events affecting the Midwest, real-time prices spiked above \$1,000/MWh for 12+ hours, generating over \$50,000 in DR revenue for a single large industrial facility with 5 MW of dispatchable DR capacity.

A Virtual Power Plant aggregates diverse DERs — rooftop solar, battery storage, smart thermostats, EV chargers, commercial HVAC, and industrial loads — into a unified, dispatchable resource capable of participating in wholesale power markets. The VPP operator's platform continuously monitors asset availability, state of charge, occupant constraints, and grid conditions to determine dispatch feasibility and optimize portfolio response.

The VPP dispatch workflow consists of: (1) asset registration and baseline characterization (measuring normal load profiles for M&V); (2) event forecast and pre-positioning (pre-charging batteries, pre-cooling buildings before events); (3) automated dispatch execution (sending control signals via OpenADR, direct API, or DERMS); (4) real-time monitoring and re-dispatch (adjusting signals based on performance vs. target); and (5) settlement and reporting (calculating measured baseline deviation for payment). VPP platforms such as AutoGrid, Enel X, and CPower manage portfolios of 500 MW–2 GW across thousands of assets.

Demand response performance measurement requires a rigorous baseline methodology to determine actual load reduction achieved during DR events. The most widely used baseline approach is the Customer Baseline Load (CBL), calculated as the average of the 10 highest load days in the preceding 45 days, excluding event days, with same-day adjustments allowed in some market rules. Baseline accuracy is critical: a 10% baseline error on a 1 MW DR resource translates to \$10,000–\$50,000 annual settlement error in typical market structures.

Pre-positioning is the practice of anticipating DR events and modifying building conditions before the event begins, creating a 'thermal battery' in building mass or storage systems. For a DR shed strategy relying on HVAC setpoint elevation (raising cooling setpoints from 72°F to 76°F during 2-hour event), pre-cooling the building to 69–70°F in the 2 hours before the event creates a buffer that maintains acceptable comfort through the full event period while achieving the full demand reduction commitment.

AI-based pre-positioning algorithms analyze weather forecasts, occupancy predictions, thermal mass characteristics, and real-time comfort metrics to determine optimal pre-positioning magnitude and timing. Studies at Lawrence Berkeley National Laboratory found that AI-optimized pre-positioning reduced occupant thermal comfort complaints by 65% versus naive setpoint changes, while maintaining equivalent demand reduction performance. Pre-positioning is particularly critical for events during extreme weather (>95°F outdoor) where rapid temperature rebound without pre-cooling would cause comfort violations within 30 minutes.

Transactive energy systems use price signals that reflect real-time system conditions to automatically coordinate flexible loads, storage, and distributed generation. Unlike conventional DR programs with fixed incentive structures, transactive energy enables continuous optimization against dynamic prices. Buildings equipped with automated energy management systems can respond to day-ahead locational marginal prices (LMPs), real-time prices, or distribution-level locational marginal prices (DLMPs) to shift flexible loads to low-price periods.

A commercial building with 500 kW of flexible HVAC load, 200 kWh of battery storage, and participation in a real-time pricing program can achieve annual savings of \$35,000–\$85,000 through price-responsive optimization versus flat-rate tariffs, depending on price volatility and load flexibility. The Pacific Northwest Smart Grid Demonstration Project showed that automated price response reduced peak demand by 15% and shifted 17% of daily load to off-peak periods for participating commercial customers, generating average annual savings of \$12,000 per commercial site.

Pacific Gas & Electric's SmartAC program enrolled over 500,000 residential and small commercial air conditioners in a direct load control demand response program. During Summer 2022 heat emergencies (outdoor temperatures exceeding 110°F in the Central Valley), PG&E dispatched SmartAC events on 12 separate days, achieving peak demand reductions of 450–620 MW per event. Customer thermal comfort was maintained through intelligent duty cycling algorithms that limited any individual thermostat from rising more than 4°F above setpoint. Annual customer incentive payments averaged \$40–\$75/year per enrolled unit, with program cost-effectiveness of \$85/kW of avoided peak capacity — compared to \$180–\$350/kW for new peaker plant construction.

An automotive manufacturing complex in Ohio enrolled 35 MW of curtailable industrial load in PJM's Emergency Load Response program through aggregator CPower. The facility implemented automated demand response capability across aluminum smelting furnaces, compressed air systems, and HVAC through an OpenADR-compatible energy management system. During the January 2022 winter storm event, PJM dispatched an emergency DR event for 6 hours. The facility delivered 33 MW of actual curtailment (94.3% performance ratio), generating \$182,000 in emergency event payments. Annual capacity market revenue averaged \$420,000/year for the enrolled DR capacity.

Enel X (now Enel North America) operates one of the largest commercial VPP portfolios in North America, managing 2.8 GW of DR resources across 5,000+ commercial and industrial customers in all major U.S. RTOs. Their JuiceNet platform aggregates EV charging (650 MW), C&I; HVAC (900 MW), industrial processes (800 MW), and battery storage (450 MW). During 2023, the VPP platform dispatched over 1,400 DR events with an average performance ratio of 91.2%. Total customer payments exceeded \$145 million annually. The platform's AI dispatch algorithm reduces energy acquisition costs by 18% compared to rule-based dispatch by optimizing across all assets considering weather, market prices, and asset state simultaneously.

Baseline integrity is the foundation of credible demand response programs. Organizations should implement sub-metering at the circuit level for DR-enrolled assets to enable accurate measurement and verification. Interval data from smart meters should be validated against BMS point data and reconciled monthly. Facilities should maintain baseline load profiles updated every 30 days using rolling-average methodology.

Communication redundancy is non-negotiable for high-value DR assets. Facilities should implement dual-path communication (primary: Ethernet/fiber; secondary: cellular LTE) for all OpenADR-connected energy management systems. Communication system testing should occur

monthly, with full dispatch simulation (test events) conducted quarterly. Signal delivery confirmation receipts should be logged and reviewed for every event.

Occupant engagement programs improve DR acceptance and reduce opt-out rates. Advance notification (email/SMS 2 hours before events), real-time building display showing DR event status, and post-event summary reports showing energy savings and incentive earnings consistently increase enrollment retention by 20–35% in commercial programs. Facilities that co-design DR strategies with occupants report 40% fewer comfort complaints during events than those implementing DR without occupant communication.

Vehicle-to-Grid (V2G) technology enables bidirectional power flow between EV batteries and the grid, transforming EV fleets into distributed storage assets capable of providing frequency regulation, peak shaving, and emergency backup power. With 6 million EVs in the U.S. as of 2024 (projected 50 million by 2030), the collective battery capacity of the national EV fleet — estimated at 600 GWh — exceeds all grid-connected stationary storage by 10x. V2G-capable chargers (SAE J3068, ISO 15118) combined with utility programs could unlock 50–100 GW of grid services capacity.

Nissan's V2G pilot with enel X in Denmark enrolled 10,000 Leaf vehicles in frequency regulation markets, achieving earnings of \$800–\$1,500 per vehicle per year with measured battery degradation below 1% additional capacity loss per year compared to conventional charging. Ford's Intelligent Backup Power system (using F-150 Lightning V2G capability) provides whole-home backup at 9.6 kW continuous for 3 days, while simultaneously participating in utility DR programs. Commercial V2G deployment faces barriers including higher charger costs (\$8,000–\$15,000 vs. \$1,500 for Level 2), limited V2G-capable vehicle models, and nascent utility interconnection standards.

Edge AI — machine learning inference deployed on smart meters, smart inverters, and building controllers rather than centralized cloud platforms — is emerging as a critical enabling technology for millisecond-level grid response. Qualcomm and NVIDIA have developed edge AI processors achieving 4–20 TOPS (tera operations per second) at sub-10 watt power consumption, enabling sophisticated prediction and control algorithms at the device level. Grid edge AI enables sub-second frequency response from distributed assets, reducing the need for centralized dispatch and communication infrastructure.

Smart grid and demand response technologies represent a transformative opportunity for commercial and industrial energy consumers to transition from passive load centers to active grid participants. The convergence of AMI infrastructure, OpenADR automation, AI-based control platforms, and wholesale market access enables sophisticated demand flexibility strategies that simultaneously reduce energy costs, generate new revenue streams, support grid reliability, and advance decarbonization objectives.

Organizations that proactively develop demand flexibility capabilities — through intelligent HVAC control, battery storage deployment, EV smart charging, and VPP participation — will capture \$0.15–\$0.55/sq ft in annual value while positioning themselves as strategic partners to utilities managing the energy transition. The evolution toward transactive energy and V2G-enabled virtual power plants will further expand the value proposition for grid-interactive buildings through the 2030s.